Structured questions

|!|BCELA0370000001|!| ★★ **STSE** **Cross topic**

CCEA GCE AL Biology Assessment Unit A2 1 Jan 2013 Q3

Many human activities harm the environment. However, there is increasing awareness of the harm we cause. Legislation and the promotion of environmentally-friendly activities are approaches taken by governments to help protect the environment.

(a) The European Nitrates Directive aims to reduce nitrate enrichment of our waterways. One aspect of the Directive is to restrict the application of some forms of fertilizer on agricultural land during the winter months. Suggest how this reduces nitrate enrichment of waterways. (2 marks)

(b) Many governments promote the use of managed reed bed systems.

Reed beds of the common reed (*Phragmites australis*) can help reduce organic waste in waterways by breaking down organic compounds and then utilizing the breakdown products for growth.

An important adaptation in *Phragmites* is that oxygen can be transferred from the leaves, through the stem and into the roots. Diffusion of oxygen out of the roots creates an oxygen-rich environment in the area immediately surrounding the roots. Consequentially oxygen gradients are created over very short distances resulting in localized aerobic and anaerobic conditions–conditions that favour highly diverse microbial populations.

Investigations have shown that constructed reed bed systems are very effective small-scale biofilters (filters of biological materials), particularly in reducing nitrate levels.

(i) With reference to the nitrogen cycle, explain why localized gradients in oxygen levels are important in the breakdown and removal of organic waste. (3 marks)

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(ii) Explain the significance of the reeds 'utilizing the breakdown products for growth' in their role as a biofilter. (1 mark)

(iii) Suggest ***two*** reasons why many governments promote the use of managed reed bed systems as a means of reducing organic waste in rural areas. (2 marks)

(c) Legislation is also in place to reduce ozone depletion of the atmosphere. Ozone gas forms a layer in the stratosphere that protects against the harmful effects of ultraviolet radiation. Explain ***one*** way in which ozone depletion can harm humans. (1 mark)

##

(a) Any ***two*** of the following: (2)
- In winter plants/crops not growing/too cold for growth/day length too short for growth
- Fertilizer will not be used/absorbed by plants
- Excessive rain/ground frozen in winter would lead to more run-off
- Artificial fertilizer more soluble

(b) (i) Any ***three*** of the following: (3)
- Decomposers (bacteria/fungi) require oxygen for respiration/release of ammonia
- Nitrification/nitrifying bacteria require(s) oxygen
- Ammonia converted to nitrate
- Denitrification reduces nitrate levels/converts nitrates to nitrogen
- Denitrification is an anaerobic process
- Oxygen required for active uptake of nitrate

(ii) Removes nitrate (ammonia) from the environment/prevents nitrate causing eutrophication (algal blooms) (1)

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(iii)Any ***two*** of the following: (2)

- More cost-effective (e.g. than extending sewage systems to some rural areas/cheaper running costs qualified)
- Deals effectively with lightly contaminated outflow from septic tanks/farms
- Creates habitats/increases biodiversity
- Aesthetic reasons/country crafts (e.g. making wicker baskets)
- Self-maintaining

(8 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	2
(b)(i)	Applying concepts	3
(b)(ii)	Applying concepts	1
(b)(iii)	Critical thinking	2

|!|BCELA0370000002|!| ★★ **STSE** **Cross topic**

CCEA GCE AL Biology Assessment Unit A2 1 Jun 2012 Q8(a)(iii),(b)–(c)

(a) The use of organic fertilizer (farmyard manure), as opposed to artificial fertilizer, helps improve soil crumb (團粒) structure. Explain, as fully as possible, ***two*** distinct ways in which an improved soil crumb structure can improve crop yield.

(2 marks)

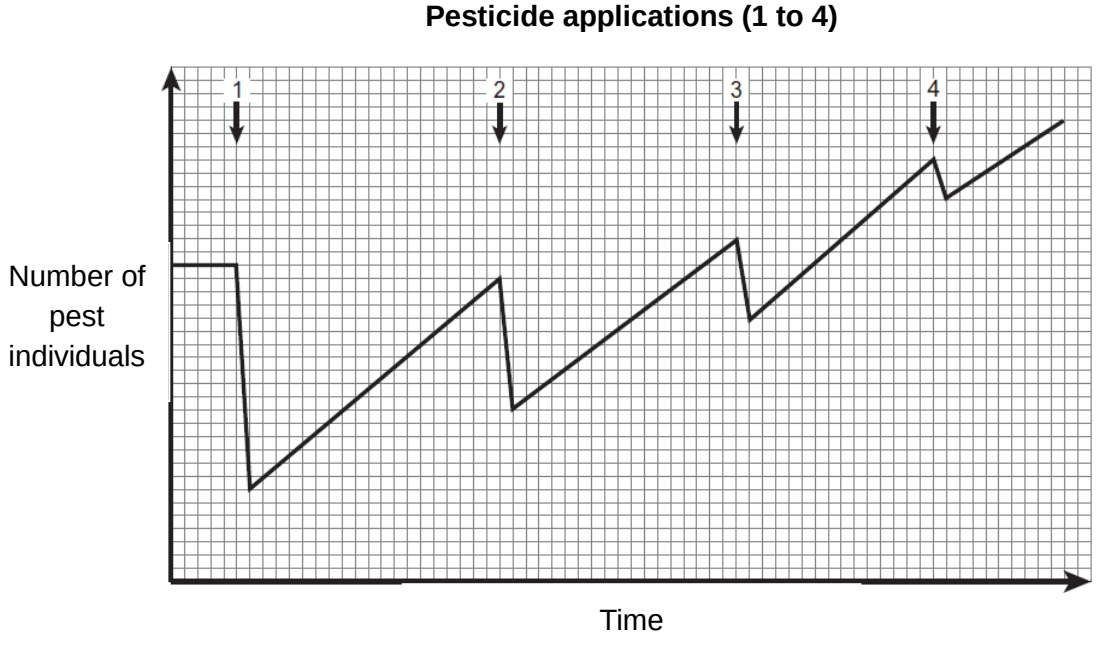
(b) Maize and other crops are subject to attack by pests.

(i) Describe and explain ***two*** ways in which pests can reduce crop yield.

(2 marks)

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Many pesticides kill a wide range of insects. The graph below shows the effect of a series of pesticide applications on the numbers of a common pest of maize plants.



(ii) Describe and explain the trends evident in the graph. (4 marks)

Integrated Pest Management Systems involve the use of a range of strategies to combat pests.

(c) Explain how the use of crop rotation and the sterilization of the males of pest species can reduce the damage caused by pests. (2 marks)

##

(a) Any ***two*** of the following: (2)

- Increased aeration for roots therefore more respiration/more active uptake of minerals
- Increased aeration for more nitrification/nitrogen fixation
- Increased drainage discourages denitrification/prevents waterlogged conditions

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that would encourage denitrification
- Increased aeration for decomposition

(b) (i) Any ***two*** of the following: (2)
- Pests can feed on/damage leaf material therefore reducing photosynthesis/growth
- They can feed on/damage the 'crop', e.g. potato
- They can feed on/damage the roots, restricting the uptake of mineral ions needed for growth
- They can spread organisms that cause crop diseases
- They can feed on the nutrients in the phloem therefore reducing carbohydrate available for storage
- Production of faeces/other waste may make the crop unsuitable for consumption

(ii) Successive application of pesticide kills fewer pests (1)
The overall number of pests increases with time (1)
Any ***two*** of the following: (2)
- Some pests survive as they did not come in contact with pesticide
- Some pests have resistance to the pesticide (and therefore survive pesticide applications)
- Resistant pests reproduce
- The pesticide also kills natural predators of the pest (causes pest resurgence)

(c) Crop rotation: Pests do not get an opportunity to complete life cycle/target crop is removed so pest numbers do not reach high numbers/different crops have different pests so no one pest can become established (1)
Sterilization of the males of pest species: Many females will fail to become fertilized therefore reducing the numbers of pests in the next generation (1)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	2
(b)(i)	Applying concepts	2
(b)(ii)	Analysing data	2
	Applying concepts	2
(c)	Understanding basic concepts	2

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|!|BCELA0370000003|!| ★★ **STSE** **Cross topic**

A survey in 2014 revealed that about 20% of the farmland in China was contaminated by heavy metals such as lead and cadmium. In an investigation, the accumulation of cadmium in a type of fern was studied. Young ferns were transplanted to soil contaminated with cadmium and were grown for 28 weeks. The table below shows the total amount of cadmium accumulated by the fern, expressed as a concentration in the plant tissue and as a percentage of the cadmium originally in the soil.

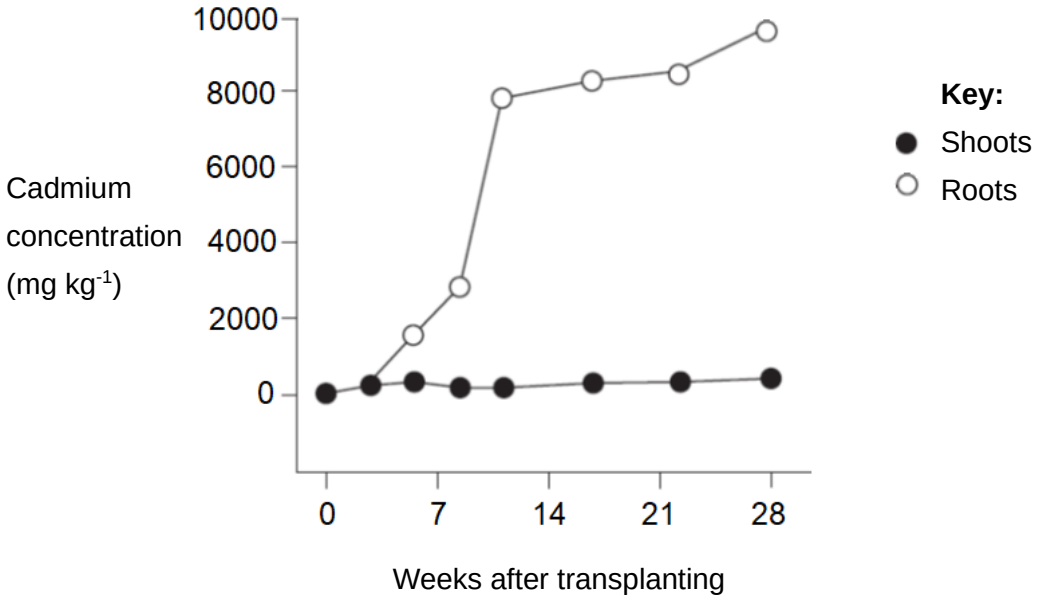
Time (weeks)	**Cadmium concentration in the fern (mg kg^{-1})**	**Percentage of original cadmium taken up by the fern**
0	0.5	0.0
7	854	1.1
14	8260	10.6
21	8710	11.2
28	13 050	16.8

(a) Suggest ***two*** possible consequences of the presence of cadmium in soil for the organisms in the community. (4 marks)

(b) Assume that the rate of cadmium accumulation in the fern over the first 28 weeks continued. Calculate how long it would take for the fern to remove all the cadmium from the soil. Show your working. (2 marks)

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The graph below shows the cadmium concentrations in roots and shoots of some of the fern during the 28 weeks after transplanting.



(c) In which part does the fern accumulate most cadmium? (1 mark)

(d) With reference to the data in the graph and the table, discuss the advantages and drawbacks of using the fern to remove cadmium from contaminated soil. (4 marks)

##

(a) Plants grown in the cadmium-contaminated soil would accumulate cadmium in their bodies. (1)
Some of them may eventually die due to the high levels of cadmium accumulated. This reduces the biodiversity of the community. (1)
As plant-grazing animals / herbivores feed on large numbers of cadmium-contaminated plants and carnivores feed on large numbers of herbivores, (1)

Cadmium can pass along the food chains and become more concentrated at each higher trophic level. It may reach a very high level that harms or kills top consumers. (1)

(Or other acceptable answers)

(b) Percentage of cadmium removed by the fern from the soil per week
= 16.8 ÷ 28 = 0.6% (1)

Time needed for the complete removal of cadmium from the soil
= 100 ÷ 0.6 = 167 weeks (1)

(c) Roots (1)

(d) Advantage:

A lower concentration of cadmium in shoots (1)

may cause less health problem to grazing animals / herbivores that consume shoots / consumers along the food chain. (1)

Disadvantage:

The removal process of cadmium is slow (1)

as only 17% of the cadmium had been removed from the soil after 28 weeks / the complete removal of cadmium by the fern takes about 167 weeks. (1)

(Or other acceptable answers)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	4
(b)	Mathematical skills	2
(c)	Analysing data	1
(d)	Analysing data	2
	Critical thinking	2

|!|BCELB0370000004|!| ★ **STSE**

CCEA GCE AL Biology Assessment Unit A2 1 Jun 2010 Q4

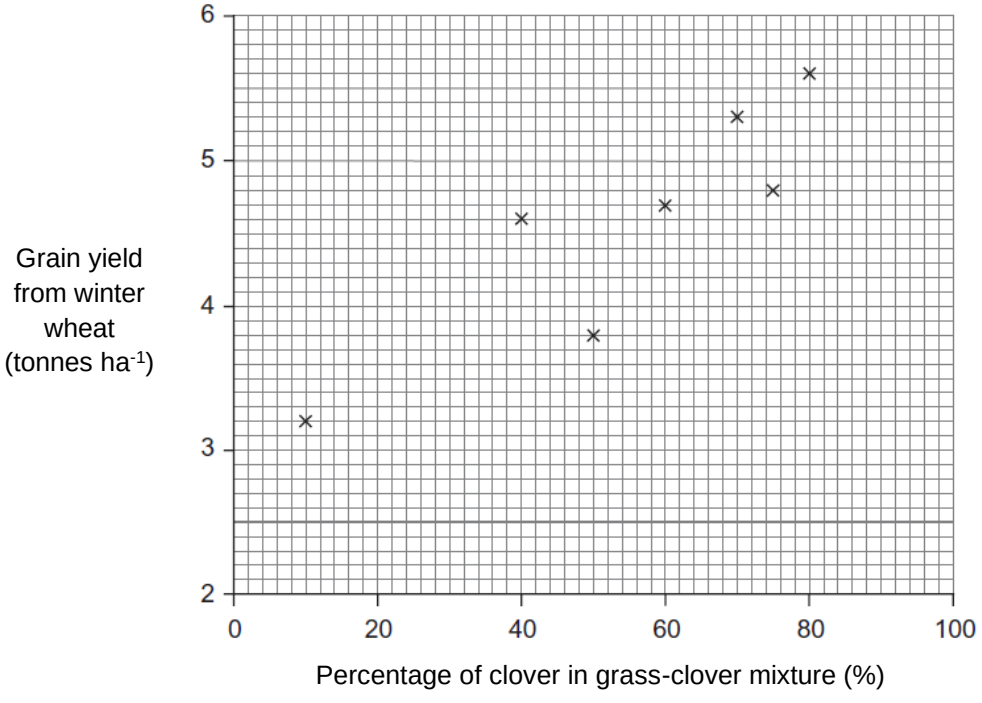
Crop rotation is used by farmers to maintain the fertility of the soil. Often crop rotation includes the planting of a grass-clover mixture. Clover is a legume with root nodules rich in amino acids and protein. In the autumn the grass-clover crop is ploughed into the soil as 'green manure' before the planting of winter wheat.

(a) (i) Explain why clover root nodules are rich in amino acids and protein.

(2 marks)

(ii) Following the ploughing of the grass-clover crop into the soil, the plants die and the soil subsequently becomes rich in nitrates. Describe the processes involved in the soil becoming rich in nitrates. (3 marks)

(b) The graph below shows the influence of different mixtures of grass-clover on the yield of wheat grain following the ploughing-in of the grass-clover crop.



(i) Explain the trend evident in the graph. (2 marks)

(ii) Describe and explain ***one*** additional benefit of crop rotation. (2 marks)

##

(a) (i) The root nodules contain nitrogen-fixing bacteria, (1)
which fix atmospheric nitrogen into amino acids. (1)
(ii) The organic nitrogen compounds in plants are decomposed, (1)
released as ammonia/ammonium compounds. (1)
Ammonia is converted by nitrifying bacteria to nitrates. (1)

(b) (i) Any ***two*** of the following: (2)
- More root nodules (with increased amount of clover in the grass-clover mix)
- Release of more ammonia/nitrates
- Production of more amino acids/proteins (which increase the crop yield)

(ii) Any ***one*** set of the following: (2)
- Pest control
the pest of a crop cannot become established when there is a gap of several years between planting of the same crop/pest life cycle broken
- Reduced competition from weeds
using different varieties of crops changes the competition for light, water and minerals
- Soil structure is improved
ploughing in of green manure/leaving cereal roots in soil (or other acceptable examples)
- Nutrient exhaustion is avoided/fewer fertilizers needed
as different crops have different nutrient requirements
- Biodiversity increased
due to reduced use of pesticides/fertilizers/variety of plants is encouraged

(9 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	2
(a)(ii)	Understanding basic concepts	3
(b)(i)	Analysing data	1
	Applying concepts	1
(b)(ii)	Understanding basic concepts	2

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|!|BCELC0370000005|!| **STSE**

Read the article below and answer the questions that follow.

> Alien species are species introduced outside their natural distribution. Without the predators or competitors in their original habitats, some of them become invasive in the introduced regions, threatening the survival of native species and disrupting the proper functioning of ecosystems.
>
> For example, the Sabah Giant Grouper, a fast-growing carnivorous fish, has been released to Hong Kong waters over the past few years. The population of this non-native species is increasing drastically and it is currently threatening other marine populations in Hong Kong waters. It is believed that the fish is released into the sea for religious reasons.

(a) Suggest reasons why the Sabah Giant Grouper would pose threat to the ecosystems in Hong Kong waters. (4 marks)

(b) The World Wildlife Fund (WWF) Hong Kong has prepared a seafood guide, encouraging people to make careful choices when consuming seafood to conserve the biological resources in the ocean. Seafood are classified into three categories: Green – 'Recommended', Yellow – 'Think Twice' and Red – 'Avoid'. Consumers are advised to avoid eating those fish in the Red category since most of these fish species are now threatened or endangered.

(i) Why is it important to conserve biological resources? (2 marks)

(ii) How could using the WWF seafood guide help conserve the biological resources in the ocean? (2 marks)

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(iii) Other than using the WWF seafood guide, suggest ***two*** ways that we can adopt to help conserve biological resources in our daily lives. (2 marks)

##

(a) The fast growth rate of the fish species allows it to out-compete other native fish species for resources such as habitats and food. (1)
Also, due to the lack of natural predators in Hong Kong waters, the populations of Sabah Giant Grouper increase drastically. (1)
The fish is predatory and the large populations feed on large amounts of native animal species. (1)
The loss of native species disrupts the existing food chains and upsets the ecological balance. (1)
(Or other acceptable answers)

(b) (i) Humans rely on biological resources for food. (1)
Biological resources are important for maintaining ecological balance. Their depletion would disrupt ecological balance. (1)
(Or other acceptable answers)

(ii) Avoid eating endangered fish species may help reduce their demand, which in turn leads to a reduction in fishing pressure of these species. (1)
Facing less fishing pressure, the fish species may recover through reproduction. (1)

(iii) Recycle used papers. (1)
Replace items made of non-renewable resources with those made of renewable resources. (1)
(Or other acceptable answers)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	4
(b)(i)	Understanding basic concepts	2
(b)(ii)	Applying concepts	2
(b)(iii)	Applying concepts	2

|!|BCELB0370000006|!| ★ **SI** **STSE**

Sewage treatment produces sludge, which contains high concentrations of nitrogen compounds. After digestion and drying, sludge can be used as fertilizer.

An investigation was carried out to study the relationship between leaching and the methods of applying sludge to the soil. Sewage sludge was applied to two regions on a piece of farmland by two different methods. In one region it was injected at various points into the soil while in another region it was spread on the soil surface.

The concentrations of nitrate in the water at the soil surface were measured after different volumes of rain had fallen. The table below shows the results of the investigation.

Rainfall (mm)	**Nitrate concentration in surface soil water (mg / 100 cm^3)**	
	Sludge injected into the soil	**Sludge spread on the soil surface**
0	2.8	1.1
50	2.9	1.7
100	3.7	1.5
150	16.2	4.7
200	25.5	4.9
250	38.5	3.5

(a) Other than nitrate, suggest ***one*** nitrogen compound that may be present in digested sewage sludge. (1 mark)

_____

(b) Describe the overall relationship between rainfall and the nitrate concentration in surface soil water. (2 marks)

_____

_____

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(c) With reference to the table, what advice would you give to farmers regarding the best time and method of applying sludge to the farmland for maximum benefit? (4 marks)

(d) Suggest ***two*** precautions that should be taken in this investigation to ensure that the results are reliable. (2 marks)

(e) Compared with inorganic fertilizers, state ***two*** advantages of using sewage sludge as fertilizer. (2 marks)

##

(a) Ammonia / ammonium compounds (1)

(b) As the rainfall increases, (1)
the nitrate concentration in surface soil water increases. (1)

(c) The sludge should be spread on the soil surface (1)
as the nitrate concentration in surface soil water was significantly lower than that using the other method, meaning less nitrate had been leached from the soil. (1)
The sludge should be applied in dry seasons / when rainfall is less (1)
as the nitrate concentration in surface soil water was lower when there was less or no rainfall. (1)
(Or other acceptable answers)

(d) The amounts of sludge applied in the two areas should be the same. (1)
The area of the two regions should be the same. (1)
(Or other acceptable answers)

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(e) Any ***two*** of the following: (2)

- Reduce the amount of sludge disposed of at landfills
- Reduce the demand for inorganic fertilizer, which in turn save natural resources
- Help improve soil structure

(Or other acceptable answers)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Analysing data	2
(c)	Analysing data	2
	Applying concepts	2
(d)	Practical skills	2
(e)	Critical thinking	2

|!|BCELB0370000007|!| ★ **STSE** **Cross topic**

Read the paragraph below and answer the questions that follow.

> Road and railway construction are inevitable in the urbanization process and thus a single habitat is divided into two or more smaller fragmented areas. This phenomenon is known as habitat fragmentation. Habitat fragmentation may favour inbreeding, which poses threat to the long-term survival of the species.

(a) Suggest why habitat fragmentation may favour inbreeding. (2 marks)

(b) Explain why inbreeding may pose a threat to the long-term survival of the species. (4 marks)

(c) Other than inbreeding, suggest ***one*** way that habitat fragmentation would cause a reduction of biodiversity in a habitat. Explain your answer. (2 marks)

(d) Suggest ***one*** measure that the government can adopt to mitigate the adverse effects of habitat fragmentation. (1 mark)

##

(a) The population size of the species in the fragmented habitat decreases due to habitat fragmentation. (1)
The choice of mates becomes limited. (1)

(b) Inbreeding often results in a decrease in genetic variation in the population. (1)
With less genetic variation, the population becomes less able to adapt to environmental changes. (1)
Inbreeding increases the proportion of homozygotes in a population. (1)
Hence, defective alleles are more likely to be expressed, leading to genetic defects which threaten the survival of the offspring. (1)

(c) Any ***one*** set of the following: (2)
- Overcrowding may occur in the smaller habitat fragments.
This leads to a more intense competition for resources. Some species may be out-competed and face extinction.
- Organisms in the habitat fragments may not be able to escape when natural disasters (e.g. flooding, hill fire, etc.) occur.
(Or other acceptable answers)

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(d) Build wildlife corridors to link up the smaller habitat fragments. (1)
(Or other acceptable answers)

(9 marks)

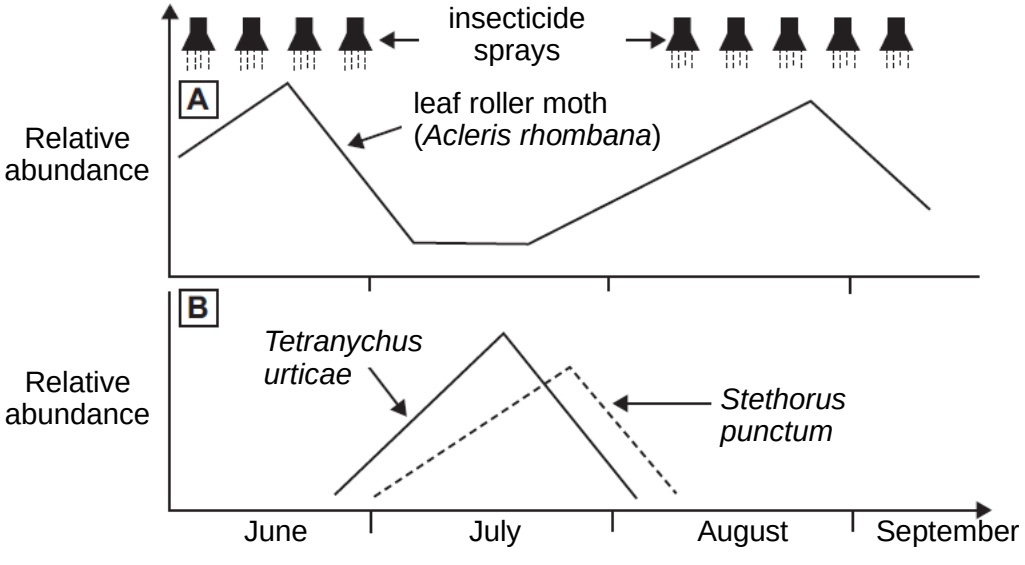
##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	1

|!|BCELA0370000008|!| ★★ **STSE** **Cross topic**

CCEA GCE AL Biology Assessment Unit A2 1 Jan 2010 Q7

Graph A below illustrates how the application of insecticides may protect an orchard (果園) by reducing the numbers of the potentially crop-damaging leaf roller moth, *Acleris rhombana*. Graph B shows how a break in the application of the insecticide protects other orchard-dwelling animals such as the mites *Tetranychus urticae* and *Stethorus punctum*.



(a) An orchard is an example of a "monoculture". Explain why monocultures are particularly susceptible to pest infestations. (1 mark)

The insecticide used in the orchard is biodegradable.

(b) (i) Identify ***one*** piece of evidence that suggests this insecticide is biodegradable. (1 mark)

(ii) Explain why the use of a biodegradable insecticide is safer for the environment. (1 mark)

Spraying of the insecticide does not take place during the month of July so as not to affect other orchard species such as the mites *Tetranychus urticae* and *Stethorus punctum*. One of these species of mite is an important pest of the fruit trees which can cause blemishing (損害) of the orchard fruit. The other species is its predator.

(c) Use the information in Graph B to help you to determine which mite species is the predator. Explain your answer. (2 marks)

By not spraying insecticide in July, the population of predatory mites remains high and so reduces blemishing of the orchard fruit by keeping the pest population low.

(d) (i) Name the type of pest control illustrated by this example. (1 mark)

(ii) Suggest ***one*** advantage of such a pest control system. (1 mark)

(iii) Suggest ***one*** possible disadvantage of this form of pest control. (1 mark)

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##

(a) Monocultures offer an enormous food source (concentrated food source) for any pest species/allow rapid colonization and spread of the pest species/no breaks in pest life cycle. (1)

(b) (i) The moth population begins to rise again (towards the end of July)/mite populations grow (during July). (1)

(ii) No bioaccumulation/no transfer to other ecosystems (e.g. soil, pond, etc.) (1)

(c) *Stethorus punctum* (1)

because its population peaks after that of its prey/as predator numbers increase prey numbers decrease. (1)

(d) (i) Biological control (1)

(ii) Any ***one*** of the following: (1)

- This method is species-specific
- Pest is controlled all year round
- Potentially harmful chemicals not sprayed on food crops
- No problems with bioaccumulation in the food chain
- When used as the sole method of pest control, crops may be sold as organic and achieve a higher price
- Increase biodiversity within the ecosystem
- After initial set up costs, this method may be cheaper

(iii) Any ***one*** of the following: (1)

- Pest is not totally eliminated (so grower has to put up with some damage to crops)
- Initial research may be long and expensive
- Biological control agent may itself become a pest
- Biological control agent may not cope with surges in pest numbers
- Biological control agent may die out and need to be re-introduced

(8 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	1
(c)	Analysing data	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Critical thinking	1
(d)(iii)	Critical thinking	1

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|!|BCELB0370000009|!| ★ **STSE**

The following photo shows a hill fire in Hong Kong countryside.



(a) State ***one*** effect of hill fires to the ecosystem. (1 mark)

(b) Describe the process of secondary succession occurring in a site that has been damaged by hill fires. (3 marks)

(c) Several decades ago, some exotic tree species were chosen to be planted in the damaged hillside in order to recover the vegetations there.

The Agriculture, Fisheries and Conservation Department (AFCD) launched the Country Park Plantation Enhancement Scheme in 2009 to plant more local tree species, instead of exotic species.

Explain the advantage of planting local tree species. (2 marks)

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(d) Besides the effort on planting tree species in the country parks, the AFCD also concerns about the conservation of some endangered plant species.
Suggest ***two*** ways in which endangered plant species can be conserved. (2 marks)

(e) Describe ***one*** problem caused by tourists visiting the country parks and suggest how to reduce this problem. (2 marks)

##

(a) Any ***one*** of the following: (1)
- Animals there are driven away or killed.
- Plants are killed by hill fire.
- Soil is not covered by vegetation and is susceptible to soil erosion.
(Or other acceptable answers)

(b) After the fire, grasses and herbs (pioneer species) start to grow on the burnt area. (1)
Some pioneer plants die and are replaced by other species (e.g. shrubs). (1)
Finally, the soil becomes so thick and rich in nutrients that it can support the growth of trees. (1)

(c) Local tree species can provide food and shelter for local animals, (1)
which in turn improves the biodiversity in the country parks. (1)

(d) Any ***two*** of the following: (1 mark each) (2)
- Artificial propagation
- Their seeds are dried and stored in a seed bank
- Legislation against international trade of the endangered plant species
(Or other acceptable answers)

(e) Any ***one*** set of the following: (1 mark for problem and 1 mark for solution) (2)

Problems	Solutions
1. Vehicles that enter the site damage the wildlife.	- access restrictions on sensitive routes
2. Human footpath erodes the soil and damage vegetation.	- access restrictions

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3. Human wastes / litters pollute the area.	- education / provision of rubbish bins / penalties
4. Wildlife may be disturbed by humans.	- public education

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)	Critical thinking	2

|!|BCELC0370000010|!| **STSE**

Read the article below and answer the questions that follow.

> 'Members of the public are reminded not to bring endangered species into Hong Kong without a licence on their return from visits to other places', a spokesman for the Agriculture, Fisheries and Conservation Department (AFCD) said.
>
> 'There has been an increase in the number of cases concerning travellers bringing back items made from endangered species without a licence. A total of 93 seizures were made at the customs checkpoints in the first six months of 2001 and the figure had reached 145 for the same period in 2002,' the spokesman said.
>
> Statistics showed that items commonly seized from travellers at customs checkpoints included orchid, cactus, American Ginseng root and crocodile products.

(a) Suggest ***two*** kinds of crocodile products. (2 marks)

(b) According to the article above, suggest ***one*** cause of extinction of species. (1 mark)

(c) Suggest ***two*** other causes of extinction of species. (2 marks)

(d) Wild orchids are endangered plants in Hong Kong. Suggest ***two*** ways to conserve these plants. (2 marks)

##

(a) Crocodile meat (1)
Crocodile leather products (1)
(Or other acceptable answers)

(b) Illegal hunting and collection of species (1)

(c) Any ***two*** of the following: (1 mark each) (2)
- Destruction of habitats of species
- Fragmentation of habitats of species
- Introduction of invasive alien species

(Or other acceptable answers)

(d) Any ***two*** of the following: (1 mark each) (2)
- Storing the seeds of wild orchids in seed banks
- Enforcement of the legislation (e.g. Protection of Endangered Species of Animals and Plants Ordinance)
- Relocation of the population of wild orchids to a suitable habitat

(Or other acceptable answers)

(7 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

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|!|BCELC0370000011|!| **Cross topic**

Figure I shows the relationship between the fishing effort and the yield of fish. The largest amount of fish that can be removed from the sea indefinitely without a decline in the stock is called the 'maximum sustainable yield'.

(Note: Fishing effort means the amount of fishing gear of a specific type used over a given unit of time)

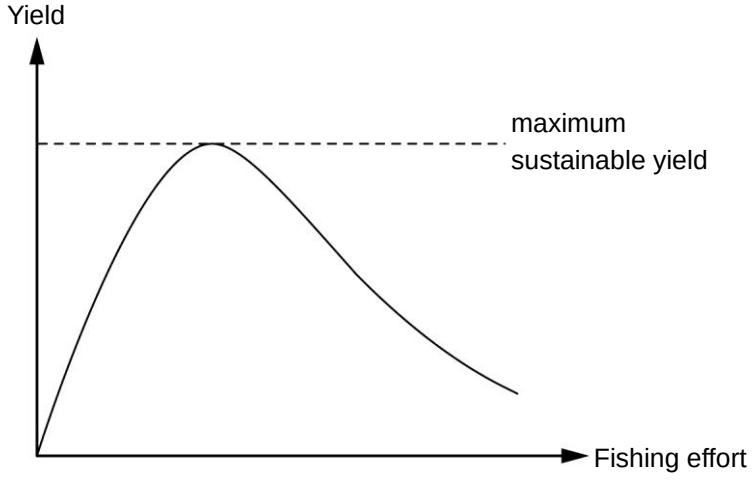


Figure I

(a) Describe the relationship between the fishing effort and the yield of fish, as shown in ***Figure I***. (2 marks)

(b) Destructive fishing methods need to be regulated in order to maintain the fish stocks.
Give ***two*** examples of destructive fishing methods. (2 marks)

(c) What is the advantage of implementing fishing moratorium? (2 marks)

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(d) ***Figure II*** shows the proportion of fish obtained from various sources.

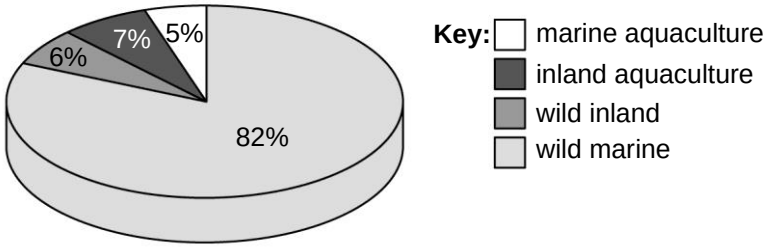


Figure II

As there is a decrease in the wild fish stocks, aquaculture becomes more important in fisheries. However, the proportion of fish obtained from aquaculture is so far relatively small.

(i) Why does aquaculture become more important? (1 mark)

(ii) State ***two*** environmental problems that may be aroused from aquaculture. (2 marks)

##

(a) The yield increases with the fishing effort until the maximum sustainable yield is reached. (1)
When the fishing effort further increases, the yield decreases as the fishing effort increases. (1)

(b) Any ***two*** of the following: (1 mark each) (2)
- Dredging
- Bottom trawling
- Cyanide fishing
- Dynamite fishing
(Or other acceptable answers)

(c) It allows fish to grow and reproduce (1)
so that the fish stocks can be replenished. (1)

(d) (i) This is because the wild fish stocks are heavily exploited, reducing the number of fish in the wild. (1)

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(ii) Any ***two*** of the following: (1 mark each) (2)

- Algal blooms
- Contamination of seawater by drugs / antibiotics used
- Escape of cultured fish into the wild will compete with / cross-breed with wild types
- Damage to habitats

(Or other acceptable answers)

(9 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Understanding basic concepts	2

|!|BCELB0370000012|!| ★

The following photograph shows a Romer's Tree Frog.



(a) Suggest ***two*** reasons to explain why Romer's Tree Frog is worth for conservation. (2 marks)

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(b) Explain how Romer's Tree Frog can be conserved by *ex situ* conservation. (3 marks)

(c) Besides *ex situ* conservation, suggest ***three*** other ways to protect and conserve the Romer's Tree Frog. (3 marks)

##

(a) It is an endemic species in Hong Kong. (1)
It is an endangered species with decreasing population. (1)
Or other reasonable answers

(b) The adults, juveniles, tadpoles and eggs of Romer's Tree Frog are collected. (1)
They are captive-bred in an environment suitable for their growth. (1)
The captive-bred individuals are then released into suitable habitats in Hong Kong. (1)

(c) Any ***three*** of the following: (1 mark each) (3)
- Setting up of conservation areas / nature reserves for the frogs
- Habitat protection
- Enforcement of legislation for protecting the frogs
- Educating the public ways of protecting the frogs
(Or other acceptable answers)

(8 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	3
(c)	Understanding basic concepts	3

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|!|BCELC0370000013|!| **STSE** **Cross topic**

(a) Suggest ***three*** reasons to explain why forest is an important resource on the Earth. (3 marks)

(b) Forest resources are renewable. However, deforestation becomes extensive and leads to many environmental problems. State ***three*** of these environmental problems. (3 marks)

(c) A land damaged by deforestation can be recovered by planting exotic or native species. Suggest ***one*** advantage of growing exotic plant species and ***one*** advantage of growing native plant species. (2 marks)

##

(a) Any ***three*** of the following: (1 mark each) (3)
- Forests carry out photosynthesis and act as carbon sink.
- Logs can be used as fuel.
- Forests provide resources of timber.
- Forests provide habitats for many species.

(Or other acceptable answers)

(b) Any ***three*** of the following: (1 mark each) (3)
- Loss of habitats
- Loss of species / biodiversity
- Reduction of fuel supplies in rural areas
- Increased soil erosion / danger of landslides
- Enhanced global warming

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(Or other acceptable answers)

(c) Any ***one*** of the following advantages of growing exotic plant species: (1)

- The species are adaptable to extreme environment / to growing on barren land.
- The species are often fast-growing.

(Or other acceptable answers)

Any ***one*** of the following advantages of growing native plant species: (1)

- The species can provide food and habitats to native animal communities.
- The species are more adapted to the local environment.

(Or other acceptable answers)

(8 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	3
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

|!|BCELC0370000014|!| **STSE**

In China, some populations of Giant Pandas are under the threat of extinction and a national nature reserve was set up in Sichuan to save the declining Giant Panda population.



(a) Suggest ***two*** factors which might lead to the extinction of Giant Pandas in the wild. (2 marks)

(b) Explain the importance of maintaining genetic diversity in the Giant Panda population. (3 marks)

(c) Suggest ***two*** ways to conserve Giant Pandas. (2 marks)

##

(a) Any ***two*** of the following: (1 mark each) (2)
- Their natural habitats are destroyed by human activities (e.g. agriculture).
- Their populations are isolated and so they breed less successfully / they have low reproductive rate.
- They are poached illegally.

(Or other acceptable answers)

(b) Genetic diversity provides variety in the phenotypes of Giant Pandas. (1)
They are more adapted to environmental changes in the wild. (1)
They have a higher chance to survive and reproduce. (1)

(c) Any ***two*** of the following: (1 mark each) (2)
- Legislation enforcement to prohibit the trading of Giant Pandas / habitat destruction
- Public education
- Captive breeding / *ex situ* conservation

(Or other acceptable answers)

(7 marks)

##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2

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|!|BCELC0370000015|!| **STSE**

Mai Po Marshes is located in the northwest New Territories. In 1976, Mai Po Marshes was designated as an SSSI. In 1995, the wetland in Mai Po and Inner Deep Bay was designated as a Ramsar site under the Ramsar Convention.

(a) What is a SSSI? (2 marks)

(b) Suggest ***two*** reasons to explain why Mai Po Marshes needs to be conserved. (2 marks)

(c) International efforts such as WWF and CITES aim at conserving ecosystems. State ***two*** ways by which WWF and CITES can benefit the ecosystems in Hong Kong. (4 marks)

(d) Suggest ***two*** limitations of CITES in the conservation of endangered species. (2 marks)

##

(a) SSSI is the Site of Special Scientific Interest. (1)
It is a conservation area where some of the most important habitats, species or geological features can be found. (1)

(b) Any ***two*** of the following: (1 mark each) (2)
- It holds many globally threatened species and some endangered species.
- It serves as over-wintering sites for many migratory birds.
- It has the largest mangrove forest in Hong Kong.

(Or other acceptable answers)

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(c) Any ***two*** ways by WWF: (1 mark each) (2)

- It manages the Mai Po Marshes.
- It promotes the use of the nature reserve for scientific research and educational purposes.
- It raises money for some environmental protection campaigns / influences the government on species protection.

(Or other acceptable answers)

Any ***two*** ways by CITES: (1 mark each) (2)

- It ensures that international trade of specimens of wild animals and plants does not threaten their survival.
- It reduces illegal wildlife trade which is the major cause of species extinction.

(Or other acceptable answers)

(d) CITES only provides protection to endangered species threatened by illegal trade. / It provides little protection to endangered species with less economic value and those species threatened by habitat loss. (1)

The enforcement efforts of CITES made by governments around the world vary / may be insufficient in some countries. (1)

(Or other acceptable answers)

(10 marks)

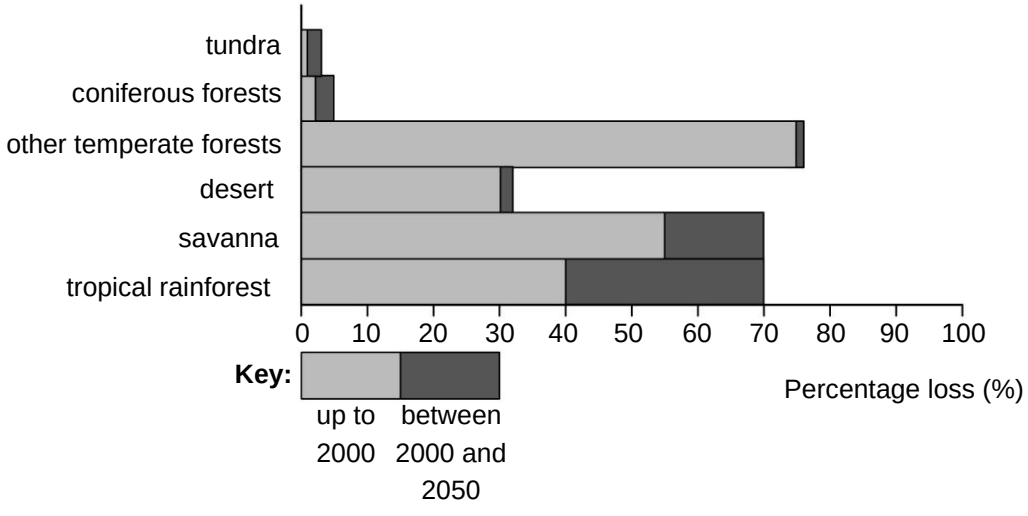
##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	4
(d)	Critical thinking	2

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|!|BCELB0370000016|!| ★ **Cross topic**

Human population has increased rapidly and food production from agriculture also increases. These lead to the overexploitation and losses of ecosystems. The following graph shows the percentage loss of ecosystems up to 2000 and predictions up to 2050.



(a) According to the graph, which ecosystem suffered the largest percentage loss up to 2000? (1 mark)

_____

(b) Up to 2000, the ecosystem losses varied from 1% to 75% for different ecosystems. Suggest ***three*** reasons to explain why percentage losses before 2000 varied so much between ecosystems. (3 marks)

_____

_____

_____

_____

_____

(c) Name the ecosystem that is expected to have the largest percentage loss between 2000 and 2050. (1 mark)

_____

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(d) Suggest ***three*** measures used for sustainable development of forests. (3 marks)

(e) Are the measures mentioned in (d) easy or difficult to carry out? Suggest ***two*** reasons to explain your views on this. (3 marks)

##

(a) Other temperate forests (1)

(b) Any ***three*** of the following: (1 mark each) (3)

- Tundra and coniferous forests are located in polar regions that are cold and unsuitable for farming and human living. Ecosystems in temperate and tropical regions have higher temperature for crops to grow.
- Deserts are hot and dry and less suitable for crops to grow. Ecosystems in savanna or forests have more rainfall for crops to grow.
- Tundra and desert are less accessible than other ecosystems and so they are less disturbed by human activities.
- Stronger economic development in temperate lands resulted in more forests to be cleared for farming / building factories / urbanization.

(Or other acceptable answers)

(c) Tropical rainforest (1)

(d) Any ***three*** of the following: (1 mark each) (3)

- Plant trees on areas with the forests cleared
- Educate local people about the sustainable ways of use of trees (e.g. avoid chopping down young trees, leave some healthy older trees to hold the soil and provide seeds for the future, etc.)
- Set up legislation against damaging of plants/habitats
- Keep forest clearances as small as possible so that the land can be regenerated quickly

(Or other acceptable answers)

(e) Usually sustainable conservation measures are not easy to implement. (1)

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Any ***two*** of the following reasons: (1 mark each) (2)

- There are often commercial, social and political pressures on the use of resources.
- Many of the remaining forests are located in developing countries which are seeking economic development.
- It increases the costs of operations / decreases profits for the companies.
- Law enforcement may be weak due to widespread corruption.

(Or other acceptable answers)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Applying concepts	3
(c)	Analysing data	1
(d)	Understanding basic concepts	3
(e)	Understanding basic concepts	3

|!|BCELC0370000017|!|

(a) The Convention on Biological Diversity was established in 1993 and one of its objectives is the conservation of biodiversity.

(i) What is biodiversity? (1 mark)

(ii) State ***two*** reasons to explain why we should conserve the biodiversity. (2 marks)

(b) Tourism will threaten habitats and hence the biodiversity.

(i) Suggest ***two*** ways in which tourism can damage habitats. (2 marks)

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(ii) Suggest ***one*** way in which ecotourism can bring benefits to the environment and ***one*** way in which it can bring benefits to the local people. (2 marks)

(c) Sustainable agriculture has become increasingly popular. Suggest ***two*** ways to achieve sustainable agriculture. (2 marks)

##

(a) (i) Biodiversity refers to the existence of a wide variety of life forms on Earth. (1)

(ii) Any ***two*** of the following: (1 mark each) (2)

- It can provide necessities to human lives.
- It can maintain the ecological balance in nature.
- It provides resources for education and scientific research.
- It should be preserved for our future generations.

(Or other acceptable answers)

(b) (i) Any ***two*** of the following: (1 mark each) (2)

- Physical destruction by construction of buildings / roads
- (Increased) pollution by waste disposal / sewage discharge
- Trampling of grassland causes harm to the grassland
- Possible degradation of habitats / wildlife for tourist entertainment

(Or other acceptable answers)

(ii) Any ***one*** of the following ways for the environment: (1)

- Fees are charged for conservation.
- Habitat conservation is encouraged by ecotourism.

(Or other acceptable answers)

Any ***one*** of the following ways for the local people: (1)

- Local people can earn money from tourists.
- It can provide jobs for the local people.

(Or other acceptable answers)

(c) Any ***two*** of the following: (1 mark each) (2)

- Practise crop rotation
- Use organic fertilizers
- Use organic pest control methods (avoid the use of pesticides)

(Or other acceptable answers)

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(9 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCELC0370000018|!| **STSE**

The following photograph shows a golden lion tamarin which is a small monkey that lives in the coastal forests of Brazil. It becomes endangered due to habitat loss.



(a) State ***two*** human activities that make golden lion tamarin an endangered species. (2 marks)

(b) The golden lion tamarin is captive-bred and then released into the wild. State ***two*** problems that may arise

(i) during the captive breeding; (2 marks)

(ii) after the release to the wild. (2 marks)

(c) Suggest ***two*** methods to manage habitats so that the plant and animal species there can be preserved. (2 marks)

##

(a) Any ***two*** of the following: (1 mark each) (2)
- Forests, where they inhabit, are cleared to provide logs.
- Forests are cleared to provide land for agriculture.
- They are captured for food / pet trade.
(Or other acceptable answers)

(b) (i) Any ***two*** of the following: (1 mark each) (2)
- Space / suitable environment for captivity is needed.
- Feeding problems may be arisen.
- Species inter-relationships need to be monitored.
- There is small gene pool / inbreeding often occurs.
- The parents lack parenting skills.
(Or other acceptable answers)

(ii) Any ***two*** of the following: (1 mark each) (2)
- The released individual may be unable to find food.
- The released individual may be unable to avoid toxic food.
- The released individual may be unable to avoid predators.
- Threat still exists in the natural habitats.
(Or other acceptable answers)

(c) Any ***two*** of the following: (1 mark each) (2)
- Prevention of habitat destruction (e.g. hill fire) that eliminates the species
- Removal of undesirable species that threatens the lives of a species
- Provision of suitable conditions that favour the growth of a species
(Or other acceptable answers)

(8 marks)

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##

	Skill assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCELB0370000019|!| ★ **STSE** **Cross topic**

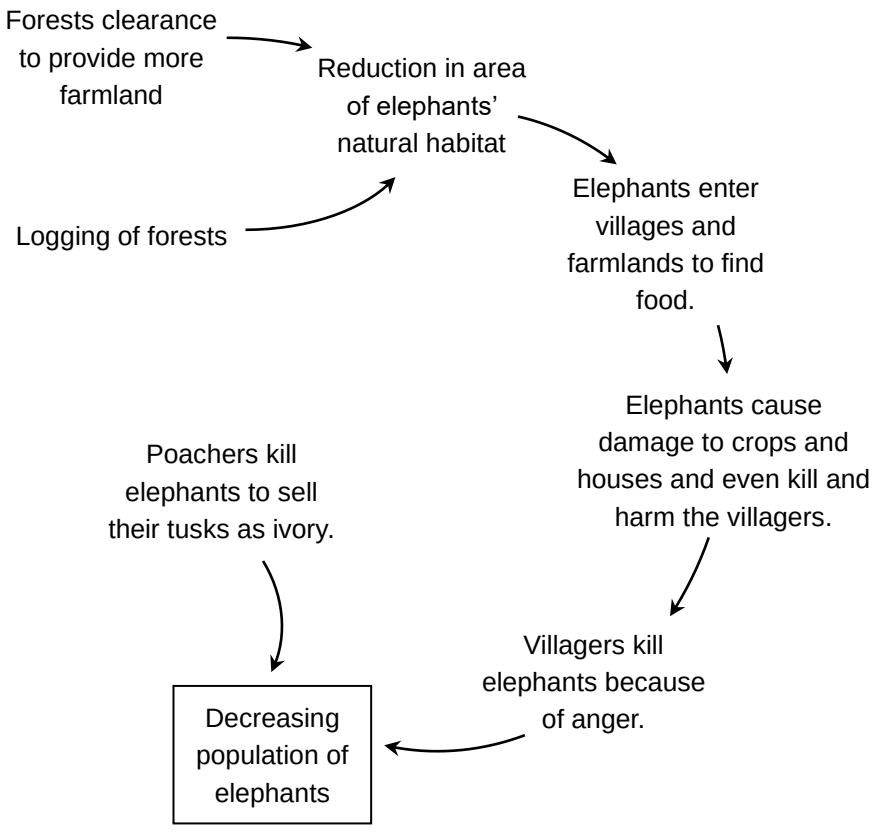
(a) The following table shows the estimated number of elephants in Sri Lanka.

Year	Estimated number of elephants in Sri Lanka
1800	33 000
1900	12 000
2000	3000

According to the data from the table, calculate the percentage decrease in the number of elephants between 1800 and 2000. Show your working. (2 marks)

(b) The diagram below shows some of the reasons that lead to a decrease in the number of elephants in Sri Lanka.

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(i) There has been a rapid increase in the human population of Sri Lanka. With reference to the diagram, explain why human population growth is likely to lead to a decrease in the elephant population. (2 marks)

(ii) Some conservationists have suggested villagers to surround villages and fields with electrified fences which forbid elephants from entering.
According to the diagram, explain how this might help conserve the elephant population. (2 marks)

(c) (i) Elephant ivory from Sri Lanka is sometimes exported to other countries as luxury products.
Name the international agreement by which most countries cooperate to ban such trade. (1 mark)

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(ii) Some academics think that if the number of elephants increases again, a small number of elephants should be allowed to be killed by licensed shooting. Explain ***one*** argument for and ***one*** argument against this point of view. (4 marks)

(iii) Describe ***one*** way in which zoos could help the conservation of animals such as elephants and other endangered species. (2 marks)

##

(a) (33000 – 3000) ÷ 33000 × 100% (1)
= 90.9% (1)

(b) (i) Growing population requires more food / shelter and hence forest is cleared as farmland / residential areas. (1)
Destruction of forests leads to loss of habitat of elephants. (1)
(ii) Electrified fences will prevent elephants from damaging crops / homes. (1)
This makes villagers less likely to kill the elephants. (1)

(c) (i) Convention on International Trade in Endangered Species of Wild Fauna and Flora (1)
(ii) Any ***one*** set of the following arguments for the view: (2 marks each) (2)
- Uncontrolled elephant population may continue to rise and increase the conflicts between villagers and elephants.
- Control of elephant population may be necessary to prevent starvation of elephants / damage to crops etc.

(Or other acceptable answers)
Any ***one*** set of the following arguments against the view: (2 marks each) (2)
- Every elephant has the right to survive.
  Humans have the responsibility to look after them / should not kill them.
- The allowance of shooting may lead to the increase in poaching and illegal

ivory trade.

This may lead to the continued reduction in the number of elephants.

(Or other acceptable answers)

(iii) Any ***one*** set of the following: (2 marks each) (2)

- Captive-breed the animals

  and re-introduce them into the wild.
- Captive animals may raise public awareness.

  This leads to fund-raising / political pressure on behalf of wild animal conservation.

(Or other acceptable answers)

(13 marks)

##

	Skill assessed	Mark(s)
(a)	Mathematical skills	2
(b)(i)	Understanding basic concepts	2
(b)(ii)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Critical thinking	4
(c)(iii)	Understanding basic concepts	2

|!|BCELB0370000020|!| ★ **STSE**

There are many old trees in Lai Chi Wo in Hong Kong. Many of them are more than 250 years old. Some people believe that old trees are important and should be conserved.

(a) What may be the danger caused by old trees? (1 mark)

_____

(b) State ***one*** reason for conserving old trees which fits each of the following descriptions.

(i) an ecological reason (1 mark)

_____

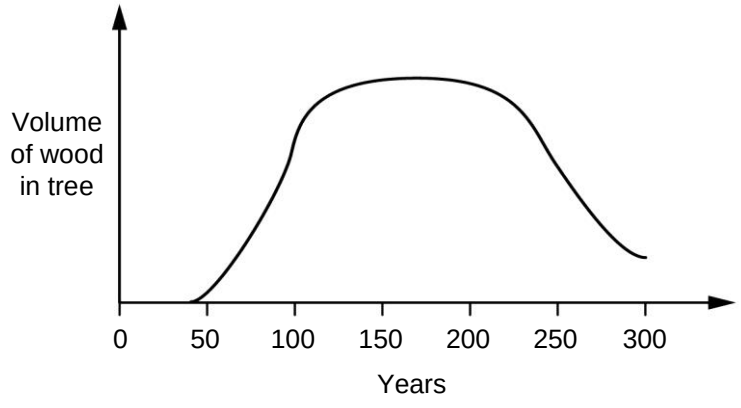
(ii) an aesthetic reason (1 mark)

_____

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(iii)an economic reason (1 mark)

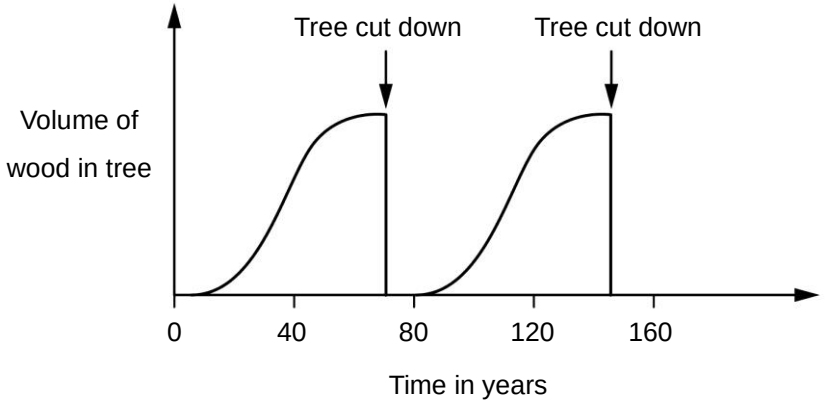
(c) ***Graph I*** shows the changes in the volume of an old tree growing in a natural woodland.



Graph I

Describe the changes in the volume of wood in the tree shown in ***Graph I*** during the first 200 years of its life. (3 marks)

(d) ***Graph II*** shows the changes in the volume of spruce trees (an exotic species) in a commercial forestry plantation.



Graph II

(i) Explain why the spruce trees in the commercial forestry plantation are cut down when they are about 70 years old. (2 marks)

(ii) Explain ***one*** argument for and ***one*** argument against commercial forestry plantation. (4 marks)

##

(a) The trees may fall (in windy season) and cause harm to human or other properties. (1)

(Or other acceptable answers)

(b) (i) Any ***one*** of the following: (1)
- Birds feed on their fruits.
- Owls and other birds nest in their holes.
- Beetles and other insects live on fallen branches.
- They are native to Hong Kong.

(Or other acceptable answers)

(ii) Many people think that old trees are beautiful / other reasonable answer. (1)

(iii) They can attract tourists / other reasonable answer. (1)

(c) From 0 to 50 years, there was very slow / no increase in its wood volume. (1)

From 50 to 100 years, there was rapid increase in its wood volume. (1)

From 100 to 200 years, its wood volume levelled off. (1)

(d) (i) The spruce trees have reached a point where their growth levels off / slows down. (1)

Therefore, it is more profitable to cut down trees and plant a new one. (1)

(ii) Any ***one*** set of the following arguments for the plantation: (2)
- It can provide habitat for some animal species.
- It can create employment and generate income.
- The roots of trees can hold soil tight and

reduce soil erosion.

- The trees can act as carbon sink and
  reduce the amount of greenhouse gas in the atmosphere.

(Or other acceptable answers)

Any ***one*** set of the following arguments against the plantation: (2)

- Exotic species are planted and
  the species support less native wildlife.
- There is usually close planting of trees and
  hence space is limited for recreational use.
- It creates monoculture
  and leads to the rapid spread of diseases / pests.
- There may be conflicts between forestry and other land uses
  such as agriculture / nature conservation.

(Or other acceptable answers)

(13 marks)

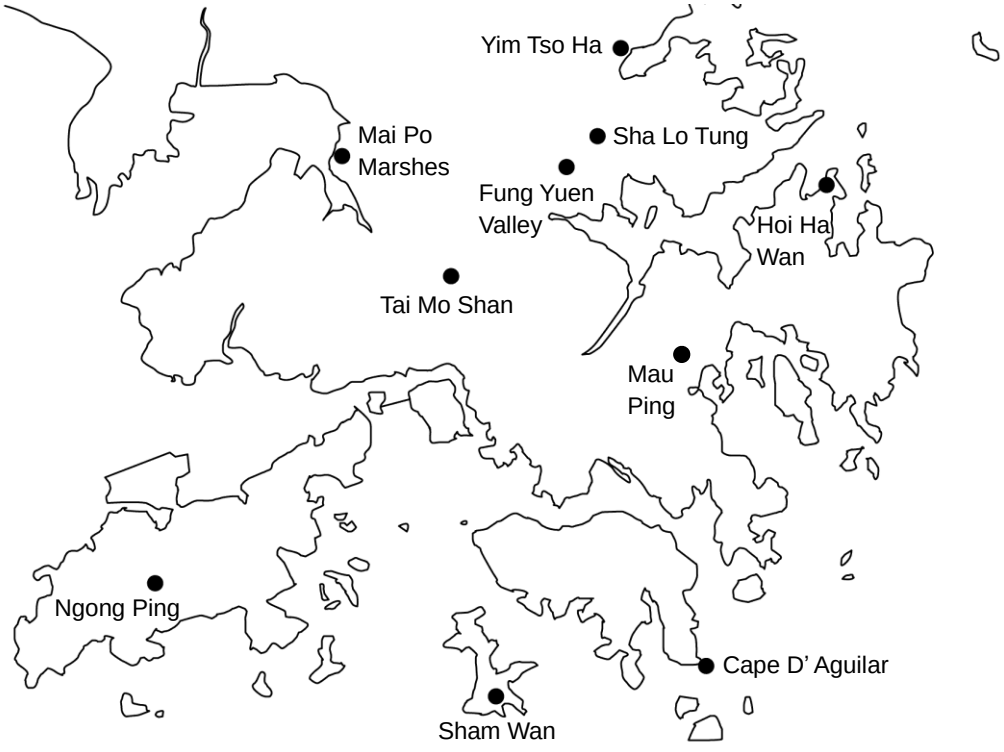
##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)(i)	Analysing data	1
(b)(ii)	Analysing data	1
(b)(iii)	Analysing data	1
(c)	Analysing data	3
(d)(i)	Applying concepts	2
(d)(ii)	Critical thinking	4

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|!|BCELB0370000021|!| ★ **STSE**

The following diagram shows the distribution of some SSSIs in Hong Kong.



(a) State ***two*** purposes of setting up SSSIs in Hong Kong. (2 marks)

(b) What special features do Mai Po Marshes, Cape D'Aguilar and Hoi Ha Wan have that lead to their designation as SSSIs? (3 marks)

(c) Hoi Ha Wan is one of the marine parks in Hong Kong. Visitors can carry out a number of recreational activities there. However, some activities are prohibited. Give ***two*** examples of these prohibited activities. (2 marks)

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(d) State ***two*** problems which may be caused by tourists visiting these SSSIs and suggest corresponding solutions to reduce these problems. (4 marks)

##

(a) To recognize the scientific importance of these sites. (1)
To ensure that consideration to conservation is given when the development in or near these sites are proposed. (1)

(b) Mai Po Marshes has the largest mangrove area in Hong Kong that has a rich diversity of aquatic plants and animals. ***OR***
Mai Po Marshes is an important resting and feeding ground for migratory birds, including some endangered species: black-faced spoonbill. (1)
Cape D'Aguilar has extensive rocky shores and subtidal areas with rare coral species. (1)
Hoi Ha Wan has diverse coral communities. (1)

(c) Unauthorized fishing (1)
Collecting of marine specimens (1)
(Or other acceptable answers)

(d) Any ***two*** sets of the following or other reasonable answers: (1 mark for a problem and 1 mark for a solution) (4 marks)

Problem	**Solution (Any one)**
1. Vehicles that enter the site damage the wildlife.	- access restrictions on sensitive routes - setting up of road charges
2. Human footpath erodes the soil and damage vegetation.	- construction of paths - access restrictions
3. Human waste / litter pollutes the area.	- education - provision of rubbish bins - penalties
4. There may be conflicts between different recreational uses of land.	- restrictions of specific activities that harm the environment - good planning on land use
5. Wildlife may be disturbed by humans.	- public education

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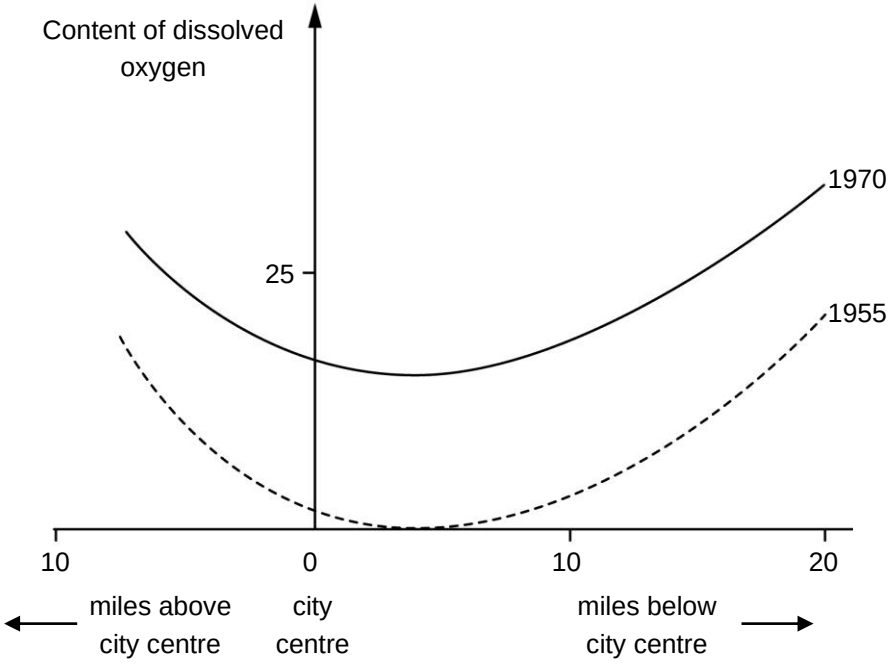
(11 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	4

|!|BCELB0370000022|!| ★

The graph below shows the changes in the dissolved oxygen content of the river water as the river flows through a city centre in 1955 and 1970.



(a) Account for the drop in the dissolved oxygen content as the river flowed into the city centre. (3 marks)

(b) Suggest ***two*** measures that led to the increase in the dissolved oxygen content in the river which occurred between 1955 and 1970. (2 marks)

(c) The water from this river is used for crop irrigation after the water has been treated by primary treatment and secondary treatment.

(i) Describe the primary and secondary treatments of waste water. (4 marks)

(ii) What further treatment is necessary before the water is discharged into the river? (1 mark)

##

(a) The domestic sewage was discharged into the river. (1)
Bacteria decomposed the organic matter in the sewage and their number increased rapidly. (1)
The great consumption of oxygen by bacteria for decomposition caused the drop in the dissolved oxygen content in the river. (1)

(b) Any ***two*** of the following: (1 mark each) (2)
- Set up sewage treatment plant
- Enforce legislation to control the discharge of sewage into the river
- Reduce the number of pollution source
(Or other acceptable answers)

(c) (i) Primary treatment:
Waste water is first screened to remove large solids and then the grit. (1)
The suspended solids in the sewage then settle at the bottom and are removed as sludge. (1)
Secondary treatment:
Microorganisms break down the organic matter into inorganic compounds. (1)
The sewage and activated sludge are separated in the final sedimentation tank. (1)

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(ii) Chlorination / disinfection (1)

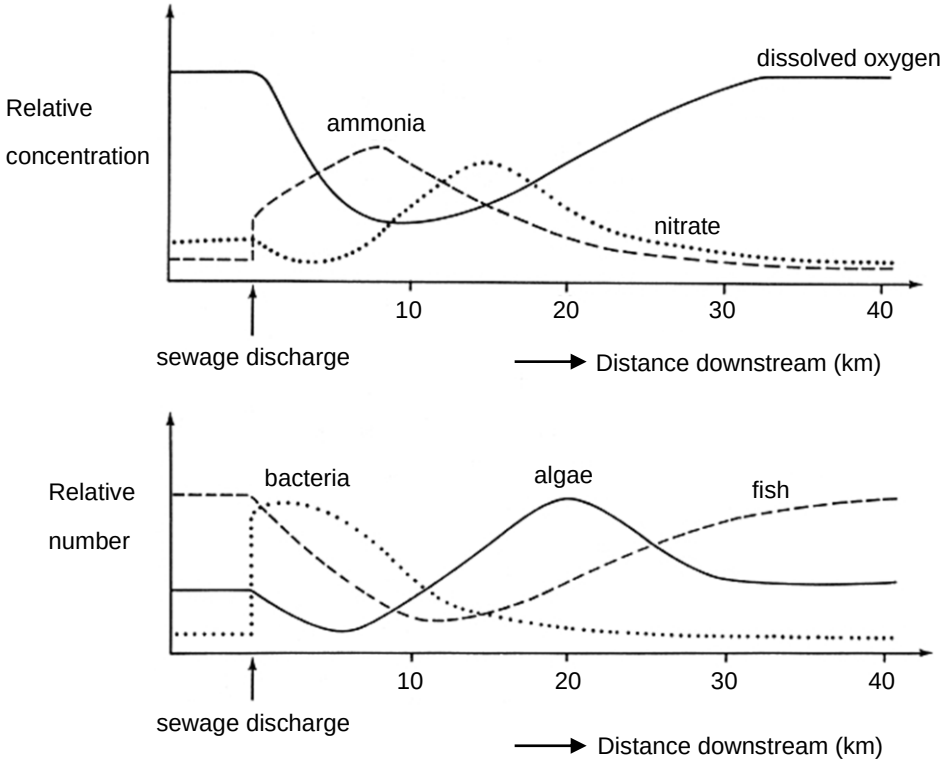
(10 marks)

##

	Skill assessed	Mark(s)
(a)	Applying concepts	3
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	4
(c)(ii)	Understanding basic concepts	1

|!|BCELB0370000023|!| ★ **Cross topic**

Sewage rich in organic pollutants was discharged into an unpolluted river. Samples of water were taken at various distances downstream along the river and were analysed for the amount of different substances and organisms. The results are shown in the graphs below.



(a) Suggest ***one*** reason to explain why the number of bacteria increased rapidly at the site of sewage discharge. (1 mark)

(b) Explain why the dissolved oxygen dropped at the site of sewage discharge and then rose gradually downstream. (4 marks)

(c) Explain the changes in the concentrations of ammonia and nitrate downstream. (4 marks)

(d) Suggest ***two*** measures that the government can adopt to prevent the discharge of raw sewage into the river. (2 marks)

##

(a) The bacteria break down the organic matter in the sewage and thus increased in number. (1)

(b) At the site of sewage discharge, more bacteria consumed oxygen for respiration. (1)

There were fewer algae and less oxygen is released during photosynthesis. (1)

Further down the stream, the bacterial population decreased and less oxygen was consumed. (1)

More algae were present and more oxygen was released during photosynthesis. (1)

(c) Sewage may contain chemical fertilizers that contain ammonium compounds. (1)

Some ammonia might be formed as a result of the decomposition of organic matter by bacteria. (1)

Ammonium compounds could be changed to nitrate by nitrification / nitrifying bacteria, causing the concentration of nitrate to increase gradually. (1)

The concentrations of both ammonia and nitrate decreased slowly downstream due

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to the dilution effect caused by the joining of other streams with the river. (1)

(d) Any ***two*** of the following: (1 mark each) (2)

- Build more sewage treatment works and sewerage system to ensure the sewage is treated before discharge.
- Set up legislation against the discharge of untreated sewage into sensitive water bodies.
- Educate the public about the environmental consequences of the discharge of untreated sewage into water bodies.

(Or other acceptable answers)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	1
(b)	Applying concepts	4
(c)	Applying concepts	4
(d)	Applying concepts	2

|!|BCELC0370000024|!| **STSE**

The following table shows the quantities of different types of municipal solid waste recovered from 2010 to 2012.

Type of municipal solid waste recovered	Quantity (kilotonnes)		
	2010	2011	2012
Paper	1195	1278.4	1162.3
Plastics	1577	843.2	316.6
Wood	17	17.7	9.0
Ferrous metals	566	667.3	499.8
Non-ferrous metals	155	115.1	78.2
Glass	5	4.8	18.3
Food Waste	Nil	0.6	6.7

(a) State ***one*** type of solid waste that has an increase in the quantity recovered from 2010 to 2012. (1 mark)

_____

(b) The Hong Kong government has adopted the Programme on Source Separation of Domestic Waste in 2005.

Describe how the programme can facilitate the recovery of the municipal solid waste in Hong Kong. (2 marks)

(c) Suggest, with examples, ***two*** ways to reduce pollution with reference to the 4Rs concept in your daily life. (2 marks)

(d) Explain how the plastics and food wastes cause problems if they are disposed of in the landfill sites. (4 marks)

##

(a) Glass / Food waste (1)

(b) The three-coloured recycling bins can be found in many places in Hong Kong. (1)

The bins separate recyclable items by category, such as waste paper, metals and plastics. (1)

(c) Any ***two*** of the following: (1 mark each) (2)

- Reduce: bring your own bag for shopping and use fewer plastic bags
- Reuse: reuse plastic containers
- Replace: use cloths instead of paper tissues for cleaning
- Recycle: extract aluminium from waste soft drink can to produce new cans

(Or other acceptable answers)

(d) Plastics are non-degradable (1)

and it will occupy place in the landfill sites which are almost full. (1)

Food wastes can be decomposed by bacteria to give out biogas. (1)

Food wastes may leach into the sea and this will cause pollution. (1)

(9 marks)

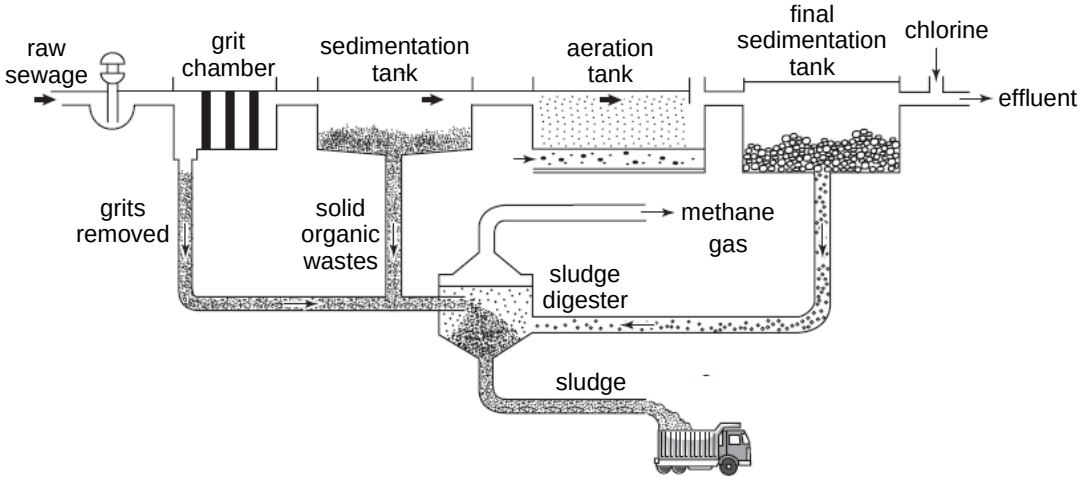
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##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	4

|!|BCELC0370000025|!|

The diagram below shows the construction plan of a sewage treatment plant for treating domestic sewage.



(a) Give the names of the physical or biological processes that take place inside the following compartments.

(i) sedimentation tank (1 mark)

(ii) aeration tank (1 mark)

(iii) sludge digester (1 mark)

(b) Explain why air is bubbled through the aeration tank. (4 marks)

(c) Describe the fate of ammonium compounds in the sewage during the sewage treatment. (2 marks)

(d) Suggest ***one*** use of sludge and methane gas after they are produced in the sewage treatment plant. (2 marks)

##

(a) (i) Sedimentation (1)
(ii) Decomposition (1)
(iii)Anaerobic respiration / fermentation (1)

(b) Bubbling air increases the oxygen supply to microorganisms in the sewage. (1)
This facilitates the aerobic respiration of these microorganisms (1)
and stimulates their growth / metabolic activities of the microorganisms, (1)
thus promoting the decomposition of organic matter in the sewage into inorganic compounds. (1)

(c) The ammonium compounds are converted to nitrates (1)
by nitrifying bacteria under aerobic conditions. (1)

(d) Sludge: used as fertilizer (1)
Methane gas: used as a clean fuel to generate electricity (1)

(11 marks)

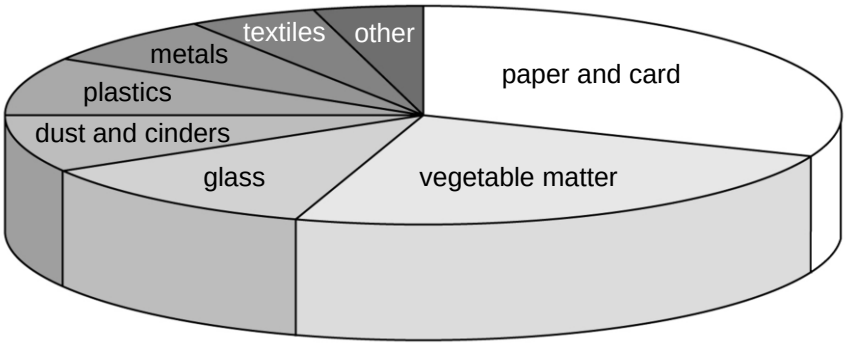
##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Understanding basic concepts	1
(b)	Understanding basic concepts	4
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2

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|!|BCELC0370000026|!|

The pie chart below shows the average composition of domestic waste from a developed city.



(a) Which of the above domestic wastes is/are not recyclable? (1 mark)

(b) (i) Suggest ***one*** reason for the large amount of vegetable matter in the domestic waste. (1 mark)

(ii) What could the citizens of the city do to reduce the amount of vegetable matter in their waste? (2 marks)

(c) State ***one*** advantage and ***one*** problem associated with the recycling of one of the wastes shown in the pie chart. (2 marks)

(d) In the city, large amounts of waste are not recycled. Instead, most are deposited in landfill sites.

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State ***two*** problems that are associated with the deposition of wastes in landfill sites. (2 marks)

(e) Suggest ***three*** ways in which the government can help reduce the domestic waste. (3 marks)

##

(a) Dust and cinders (1)

(b) (i) From food leftovers / spoiled food (1)
(Or other acceptable answers)

(ii) Order / prepare food to the right amounts. (1)
Store food properly (e.g. in a refrigerator). (1)
(Or other acceptable answers)

(c) Any ***one*** of the following advantages: (1)
- It reduces the depletion rate of natural resources.
- It reduces the amount of solid wastes generated.
- It reduces the pollution generated during the exploitation of resources.
(Or other acceptable answers)

Any ***one*** of the following problems: (1)
- It is difficult in separating different types of waste materials.
- It requires extra energy input in recycling.
(Or other acceptable answers)

(d) Any ***two*** of the following: (1 mark each) (2)
- Costly to use land as landfill site
- Shortage of landfill sites
- Pressure on land use
- Pollution by leaching
- Pollution by the production of methane
(Or other acceptable answers)

(e) Any ***three*** of the following: (1 mark each) (3)
- Encourage the separation and recycling of wastes
- Implement the charging on the use of plastic bags

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- Implement the charging on the domestic waste disposal
- Give funds to recycling industries

(Or other acceptable answers)

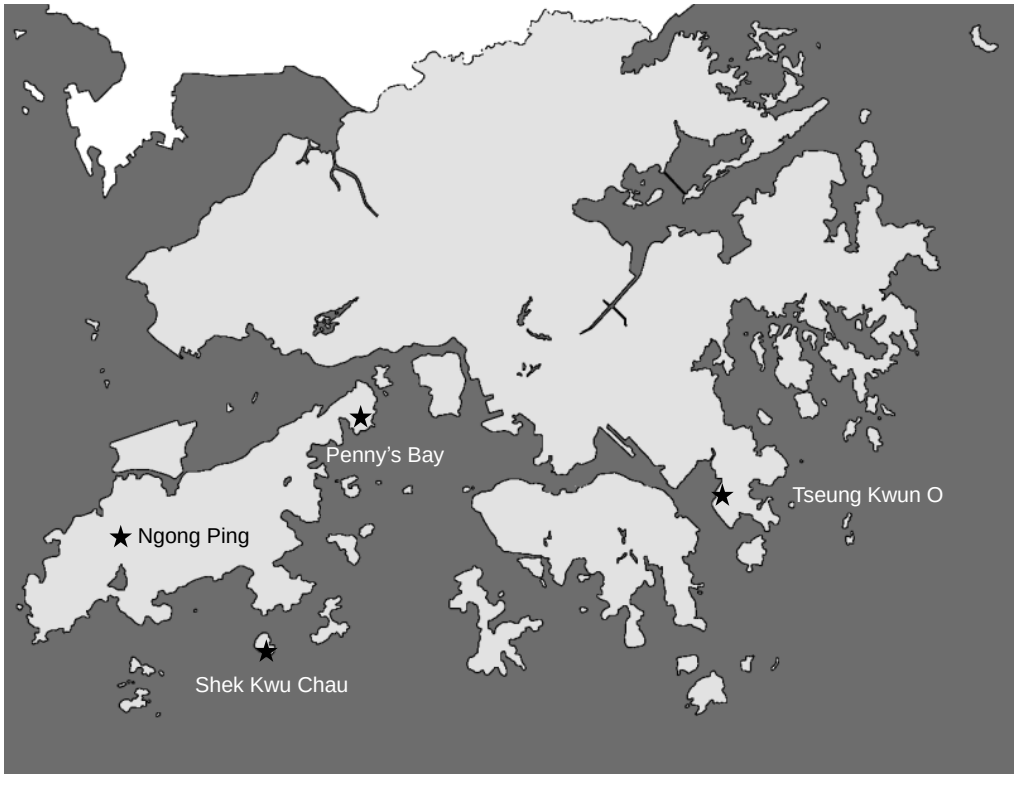
(11 marks)

##

	Skill assessed	Mark(s)
(a)	Analysing data	1
(b)(i)	Applying concepts	1
(b)(ii)	Applying concepts	2
(c)	Understanding basic concepts	2
(d)	Understanding basic concepts	2
(e)	Understanding basic concepts	3

|!|BCELB0370000027|!| ★ **STSE** **Cross topic**

The following map shows the location of some places that are concerned in pollution and pollution control in Hong Kong.



(a) Penny's Bay was reclaimed for the construction of Disneyland in the 2000s. Suggest and explain ***two*** impacts of reclamation on the environment. (5 marks)

(b) Tseung Kwun O has one of the landfill sites in Hong Kong.

(i) State ***one*** problem that is associated with the landfill sites. (1 mark)

(ii) Incineration is another method to treat waste and Shek Kwu Chau is one of the potential sites for establishing the incineration plant.
State ***one*** benefit and ***one*** drawback of treating waste with incineration. (2 marks)

(iii) Suggest ***three*** methods to reduce the amount of solid waste. (3 marks)

##

(a) Reclamation destroys natural habitats in coastal areas, (1)
resulting in a loss of biodiversity. (1)
Reclamation involves dredging (1)
Any ***two*** of the following: (1 mark each) (2)
- Dredging destroys seabed habitats. This leads to a loss of biodiversity.
- Dredging stirs up mud from the sea bottom, increasing the amount of suspended solids in water. Suspended solids may clot fish gills, causing many fish to die.
- Suspended solids prevent sunlight from penetrating into water. This reduces the primary productivity of the marine ecosystem.

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- The mud dredged from the seabed may contain toxic chemicals, which may poison marine organisms.

(Or other acceptable answers)

(b) (i) Any ***one*** of the following: (1)

- The capacity of the landfill site is limited.
- The bad smell from the landfill site may disturb people.
- Leaching may occur and the leachate contaminates underground water / causes water pollution in water bodies.
- The methane produced at landfills can cause global warming if released into the atmosphere.

(Or other acceptable answers)

(ii) Any ***one*** of the following benefits: (1)

- It can greatly reduce the amount of waste.
- The burning can release energy that can be used to generate electricity.
- Less land is needed when compared with landfill sites.

(Or other acceptable answers)

Any ***one*** of the following drawbacks: (1)

- It may emit some harmful gases during the combustion of solid waste.
- The ash produced may pollute the air.

(Or other acceptable answers)

(iii) Any ***three*** of the following: (1 mark each) (3)

- Set up environmental levy on plastic shopping bags
- Recycle the solid wastes
- Charge for solid waste disposal
- Reduce the use of materials / reuse the materials / replace the materials

(Or other acceptable answers)

(11 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	5
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	3

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|!|BCELB0370000028|!| ★ **Cross topic**

(a) (i) What is meant by monoculture? (1 mark)

(ii) Suggest ***three*** problems that may be arisen when a farmer practises monoculture. (3 marks)

(b) (i) What is meant by crop rotation? (2 marks)

(ii) Design and explain a crop rotation system that could be used to grow three types of crops (root vegetable, leafy vegetable and leguminous plant) in a field which has been divided into three parts. (4 marks)

(iii) State ***two*** benefits of practising crop rotation. (2 marks)

##

(a) (i) A single crop species is grown over a large area for a number of years. (1)

(ii) Any ***three*** of the following: (1 mark each) (3)

- Depletion of certain minerals from the soil
- Increase in pest population and rapid spread of diseases

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- Addition of more fertilizers by farmers
- More reliance on pesticides by farmers
- Disturbance of ecological balance

(Or other acceptable answers)

(b) (i) It is the practice of growing different types of crops in the same field (1)
in different seasons of the year or year by year. (1)

(ii) The crop rotation system should be leguminous plant → leafy vegetable → root vegetable → leguminous plant → … (1)

This is because a leguminous plant can enrich the soil with nitrogen. (1)

A leafy vegetable which requires lots of nutrients can grow better in nitrogen-rich soil. (1)

The following root vegetable requires fewer nutrients from soil and can grow in nutrient-depleted soil. (1)

(iii) Any ***two*** of the following: (1 mark each) (2)

- Different nutrients in the soil are used in different parts of each cycle.
- It reduces the build-up of pests / diseases / weeds.
- The soil fertility can be restored if leguminous plants are included.

(Or other acceptable answers)

(12 marks)

##

	Skill assessed	Mark(s)
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	3
(b)(i)	Understanding basic concepts	2
(b)(ii)	Applying concepts	4
(b)(iii)	Understanding basic concepts	2

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|!|BCELB0370000029|!| ★ **STSE**

The Hong Kong government constructed sewage treatment works in order to improve the water quality of the sea water.

Ngong Ping Sewage Treatment Works is the first tertiary sewage treatment works in Hong Kong.

(a) What is the importance of sewage treatment? (2 marks)

(b) Describe the primary and secondary treatments of water in Ngong Ping Sewage Treatment Works. (6 marks)

(c) In tertiary treatment, the sewage is treated to remove inorganic nutrients (e.g. nitrates and phosphates). Suggest why the sewage should be treated by tertiary treatment before discharged into sensitive waters (e.g. freshwater rivers and lakes). (2 marks)

##

(a) It can remove the solid waste and organic matter in domestic sewage. (1)
As a result, the treated water contains fewer organic pollutants and it causes less harm to the environment. (1)

(b) In primary treatment, the sewage is first treated to remove large solid wastes / suspended solids using physical processes (screening, removal of grit and sedimentation). (1)
In secondary treatment, the sewage is passed into an aeration tank with the

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bubbling of air. (1)

Bubbling air increases the oxygen supply to microorganisms in the sewage. (1)

This facilitates the aerobic respiration of these microorganisms (1)

and stimulates their growth / metabolic activities of the microorganisms, (1)

thus promoting the decomposition of organic matter in the sewage into inorganic compounds. (1)

(c) The discharge of sewage rich in inorganic nutrients (e.g. nitrates and phosphates) leads to an increase in the nutrient content in water / eutrophication. (1)

This causes algal blooms, which may result in adverse ecological consequences. (1)

(10 marks)

##

	Skill assessed	Mark(s)
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	6
(c)	Understanding basic concepts	2

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